MATHEMATICS
Introduction to Modern Algebra II
Quiz 4.

1) Define the characteristic of a field and prove that it is either 0 or a prime number.

Proof. Let F be a field and 1 its multiplicative unit. Let $S = \{t \in \mathbb{N} | t \cdot 1 = 0\}$ (here $t \cdot 1 = 1 + 1 + \ldots + 1$). If $S = \emptyset$, then the characteristic is 0, otherwise by the WOP there is a smallest element $p \in S$.

Note that $p \neq 1$ since $1 \cdot 1 = 1 \neq 0 \in F$.

If $p = uv$ where $u, v \in \mathbb{N}$, then $0 = p \cdot 1 = (uv) \cdot 1 = (u \cdot 1) \cdot (v \cdot 1)$. As F is a domain either $u \cdot 1 = 0$ or $v \cdot 1 = 0$. But then as $u, v \leq s$ either $u = p$ or $v = p$ so that p is prime. $\square$

2) Show that the vectors $(3, 1, 4)$, $(4, 3, -3)$, $(2, 2, 3)$ are linearly dependent over $\mathbb{Z}_5$ and linearly independent over $\mathbb{Z}_{11}$.

Proof. $(3, 1, 4)$, $(4, 3, -3)$, $(2, 2, 3)$ are linearly dependent if we can find x, y, z not all 0 such that $x(3, 1, 4) + y(4, 3, -3) + z(2, 2, 3) = 0$ i.e. if we have nontrivial solutions to $3x + 4y + 2z = 0$, $x + 3y + 2z = 0$, $4x - 3y + 3z = 0$. This system is equivalent to $-5y - 4z = 0$, $x + 3y + 2z = 0$, $-15y - 5z = 0$ and hence to $-5y - 4z = 0$, $x + 3y + 2z = 0$, $7z = 0$.

In $\mathbb{Z}_5$ this is $z = 0$, $x + 3y + 2z = 0$, $2z = 0$ and so $(-3, 1, 0)$ is a nontrivial solution.

In $\mathbb{Z}_{11}$, the third equation $7z = 0$ implies $z = 0$ and so from the first equation we get $-5y = 0$ and hence $y = 0$ and so from the second equation we get $x = 0$. The only solution is the trivial one. $\square$